

How Big Is the Breeze Exchanger? Sizing the Stage 1 Plate Heat Exchanger and Its Real Pressure Drop

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67 Pa calculated pressure drop (200 Pa assumed)	2.7 m stack height, 1.2 × 1.2 m plates	~8× larger than the exchangers in the CAD	1.08–1.09 mechanical PUE with sized exchanger
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1. Summary

The energy analysis showed that once the chiller is gone, fans use most of the cooling electricity, and fan power depends on how hard the air is to push through each component (its pressure drop). The Stage 1 air-to-air exchanger was assumed to cost 200 Pa on each side. This report sizes that exchanger from first principles to replace the guess with a calculated number.

Result: a plain aluminium plate exchanger with a 3 mm gap and 1.2 m plates reaches the required 75% effectiveness with only 67 Pa per side. That lowers the hybrid's cooling electricity from 275 to 218 MWh per year (188 MWh with the seawater-coil bypass), mechanical PUE 1.08–1.09. The catch is size: each of the six modules is a 1.2 × 1.2 m stack 2.7 m tall, about eight times the volume drawn in the current CAD.

2. What Is Being Sized

Stage 1 is a cross-flow plate heat exchanger, the usual type for indirect air-side economisers. It is a stack of thin aluminium plates with small gaps between them. Room air flows through every other gap in one direction; outside air flows through the gaps in between at 90°. The two streams never mix: heat passes through the metal plates.

Design requirement	Value	Source
Effectiveness	0.75 at equal room and outside airflow	Heat exchanger chain model
Airflow per module	3.35 m³/s each side (10.06 m³/s per path ÷ 3 modules)	CFD Test 1 / CAD layout
Modules	3 per path, 6 in total	CAD layout
Plate material	Aluminium, 0.15 mm	Typical for air-to-air plate exchangers

To reach 0.75 effectiveness with equal flows, a cross-flow exchanger needs NTU ≈ 5.05, which means a conductance (UA) of about 20.1 kW/K per module.

3. Method

For each combination of plate gap (3–8 mm) and plate size (0.6–1.5 m), the model finds how many plates are needed to reach that UA, then calculates the resulting stack height, volume and pressure drop.

- **Flow regime:** channel Reynolds numbers are 150–1,050, so the flow is laminar, smooth and orderly rather than turbulent.
- **Heat transfer:** fully developed laminar flow between parallel plates, $Nu = 7.54$, so $h = Nu \cdot k / Dh$ with hydraulic diameter $Dh = 2 \times \text{gap}$. In laminar flow h does not depend on air speed, only on the gap.
- **Pressure drop:** laminar friction ($f \cdot Re = 96$) along the plate plus one velocity head for entry and exit losses.
- **Plate area:** with the same h on both sides, $UA = h \cdot A / 2$, so $A = 2 \cdot UA / h$.

A useful consequence: because h depends only on the gap, the required plate area and the box volume depend only on the gap. Plate size just changes the shape: large plates make a short, wide stack with longer flow paths and more pressure drop; small plates make a tall, narrow stack with less.

4. Results

4.1 The size–pressure drop trade-off

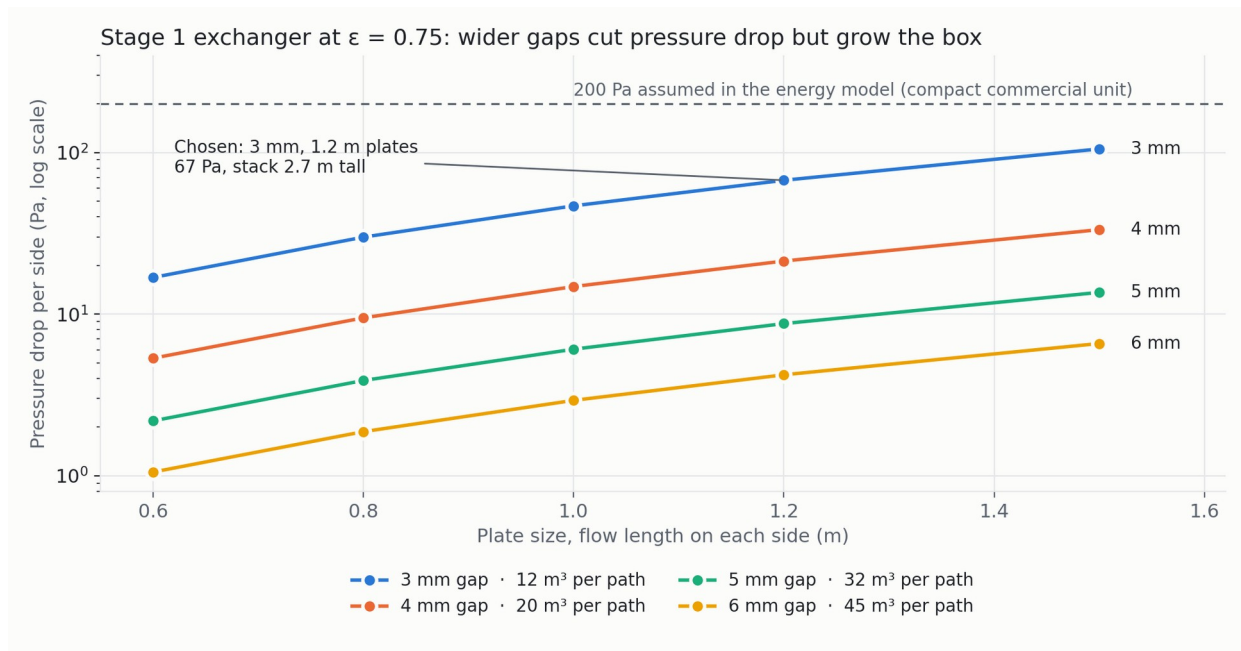


Figure 1. Pressure drop per side versus plate size for each gap, at 75% effectiveness. The dashed line is the 200 Pa assumed in the energy analysis.

Gap	Plate size	Plates per module	Stack height	Volume per path	Air speed in channels	Pressure drop
3 mm	1.0 m	1,231	3.9 m	11.6 m³	1.8 m/s	47 Pa
3 mm	1.2 m	855	2.7 m	11.6 m³	2.2 m/s	67 Pa
3 mm	1.5 m	547	1.7 m	11.6 m³	2.7 m/s	105 Pa
4 mm	1.2 m	1,140	4.7 m	20.4 m³	1.2 m/s	21 Pa

Gap	Plate size	Plates per module	Stack height	Volume per path	Air speed in channels	Pressure drop
4 mm	1.5 m	729	3.0 m	20.4 m ³	1.5 m/s	33 Pa
5 mm	1.5 m	912	4.7 m	31.7 m ³	1.0 m/s	14 Pa
6 mm	1.5 m	1,094	6.7 m	45.4 m ³	0.7 m/s	7 Pa

Every plain-plate design comes in well under the 200 Pa that was assumed. Wider gaps drop the pressure further but grow the exchanger fast: going from 3 mm to 6 mm cuts pressure drop about tenfold but quadruples the volume.

4.2 Chosen design

Chosen module: 3 mm gap, 1.2 m plates	Value
Plates per module	855 (1,230 m ² of plate area)
Module size	1.2 × 1.2 m footprint, 2.7 m tall (3.9 m ³)
All six modules	8.6 m ² of deck, 23 m ³ in total
Pressure drop	67 Pa per side
Aluminium	about 0.5 t per module, 3 t in total

The 3 mm gap is the most compact option that stays comfortably below 100 Pa and keeps the stack under 3 m. A wider 4 mm gap would tolerate more salt and dust fouling on the outdoor side, but the stack grows to 3.0–4.7 m.

4.3 Does it fit?

The current CAD shows each module as a 1.15 × 1.15 × 0.35 m box, about 0.46 m³. The sized module is about 3.9 m³, roughly eight times larger. The footprint barely changes; the exchangers need to become tall stacks instead of thin panels. Six 1.2 × 1.2 m stacks take about 8.6 m² of a deck of roughly 84 m², so they fit on the pier, but the CAD layout and ducting need to be redrawn around them.

4.4 Effect on energy

Stage 1 exchanger	Pressure drop	Hybrid cooling energy	Mech. PUE	With coil bypass
Assumed compact unit	200 Pa	275 MWh/yr	1.12	245 MWh/yr (1.10)
Plain plates, 3 mm, 1.5 m	105 Pa	234 MWh/yr	1.10	204 MWh/yr (1.09)
Plain plates, 3 mm, 1.2 m (chosen)	67 Pa	218 MWh/yr	1.09	188 MWh/yr (1.08)
Plain plates, 4 mm, 1.5 m	33 Pa	204 MWh/yr	1.09	174 MWh/yr (1.07)

With the sized exchanger, the energy premium of the breeze-first design over pumping seawater every hour (157 MWh/yr) shrinks from about 118 to 61 MWh per year, or to 31 MWh with the coil bypass. At \$0.125/kWh that is about \$3,900–7,600 a year for pumping 85% less Bay water.

5. Findings

- **The 200 Pa assumption was conservative.** Every plain-plate design tested at 75% effectiveness stayed under about 105 Pa per side.
- **The gap sets the size; the plate size sets the shape.** Tighter gaps mean smaller boxes but more pressure drop.
- **The exchangers in the CAD are about eight times too small** for 75% effectiveness with plain plates. They fit on the pier as 2.7 m stacks.
- **There is a real design choice:** large plain-plate stacks with low fan power, or compact corrugated units (typical of commercial products) that fit easily but cost roughly 150–250 Pa.
- **With a sized exchanger, the energy cost of protecting the Bay is small:** mechanical PUE 1.08–1.09 against 1.07 for seawater first.

6. Limitations

- The laminar heat transfer value ($Nu = 7.54$) ignores the better heat transfer near each channel inlet, so the plate area is slightly overestimated.
- Plates are assumed flat and perfectly spaced. Real plates need spacers or dimples, which add some pressure drop.
- Salt and dust fouling on the outdoor side will narrow a 3 mm gap over time; cleaning intervals and a fouling allowance are not included.
- Manifold and header losses around the stack are not included.
- The energy model still scales outdoor fan power with speed cubed; for laminar exchanger flow, part of the pressure drop scales more gently.

7. Next Steps

- Redraw the CAD with six $1.2 \times 1.2 \times 2.7$ m exchanger stacks and re-route the ducts.
- Compare against vendor data for a commercial corrugated-plate unit at the same airflow.
- Add a fouling allowance and check whether a 4 mm outdoor-side gap with a 3 mm room-side gap is a better balance.

Code and data: github.com/sharanankathi/bay-breeze-cooling (hx_sizing.py). Heat transfer relations: Incropera et al., Fundamentals of Heat and Mass Transfer; Shah & London, laminar flow in ducts.

Research idea, design decisions and modeling direction by the author. AI tools were used to help write code, run calculations and draft this report.